

Empirical Architecture

This document explains how the experiments in this repository are organized and how each experiment maps back to the theory in the paper. The empirical design is built as an evidence ladder: known-truth synthetic tests establish the controlled recovery and collapse behavior; posterior diagnostics test the C6 bridge; real-background planted experiments test whether the same recovery logic survives realistic event streams; and raw real-data fits provide label-free predictive and structural checks.

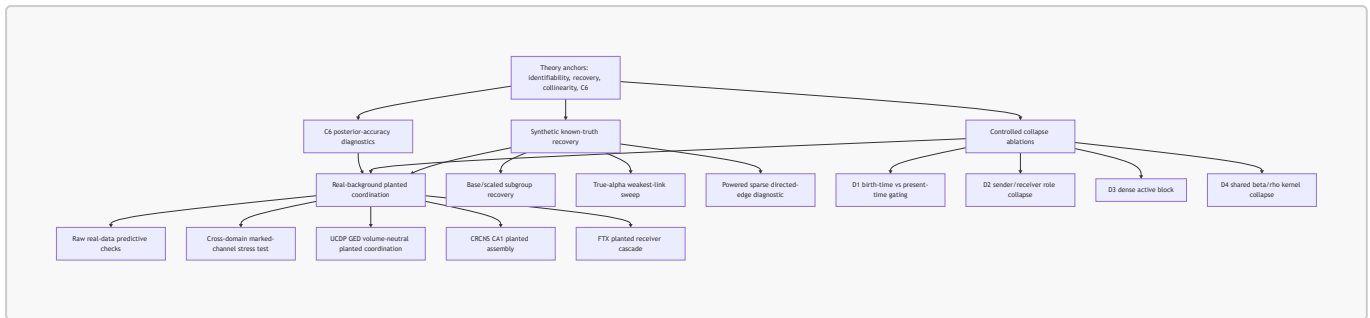
The core question we try to answer is:

Can we recover the **shape** of episodic coordination in noisy event streams: when it happened, who participated, who sent, who received, and which marked channels carried the episode?

The experiments are organized around four theorem-facing predictions. The main theory anchors are the active-kernel identifiability theorem (Theorem 1), the conditional finite-sample support-recovery theorem (Theorem 3), the birth-state-blind collinearity-collapse theorem (Theorem 2), the actor--mark scaling proposition (Proposition 1), and the local posterior-accuracy condition (Assumption Assumption 2, item 4; Condition C6 in Appendix A.3).

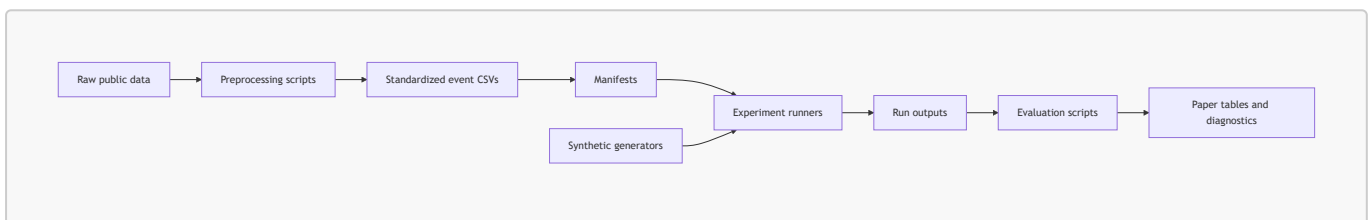
Prediction	Question	Theory anchor	Main evidence layer
P1: Active exposure	Does recovery improve with realized active-born evidence rather than total event volume?	Theorem 3: recovery is controlled by active-born effective sample size N_{eff} .	Synthetic, UCDP GED planted coordination, CRCNS CA1 planted assembly
P2: Weakest link	At fixed aggregate active mass, do weaker links require longer active sojourns?	Theorem 3: the active-duration bound scales as $1 / \alpha_{\text{min}}^2$.	Synthetic true-alpha weakest-link sweep
P3: Posterior bridge	Does recovery depend on the fitted active-state posterior being close to the oracle birth-state posterior?	Assumption Assumption 2, item 4 / Condition C6: oracle-to-approximate transfer requires local E-step posterior accuracy.	Synthetic oracle-posterior and posterior-corruption stress tests
P4: Collapse	Do the predicted failures appear when an identifiability repair is removed?	Theorem 1, Theorem 2, and Proposition 1: recovery collapses when separation, birth-state attribution, or actor--mark normalization is removed.	Synthetic ablations, CA1 segmentation-without-assignment, FTX mark-channel tests

1. High-level empirical ladder



The synthetic layer is the cleanest because the latent path, active subgroup, sender/receiver roles, marks, and true parameters are known. The real-background planted experiments preserve ground truth while replacing the background with empirical event streams. The raw real-data layer removes planted truth and is therefore interpreted only as predictive or structural evidence, not as proof of true latent coordination.

2. Repository-level data flow



All datasets are converted to the same event schema:

```
time,actor,mark
```

Some real-background planted experiments also carry extra columns such as `is_injected`, `Z`, or manifest-level truth fields. The model runner consumes the standardized event streams and manifest files; the evaluator maps fitted scores back to the planted supports, receiver targets, active windows, or mark channels depending on the experiment.

3. Synthetic known-truth recovery

Goal

Test whether the proposed regime-gated marked Hawkes estimator recovers the planted active subgroup, hub role, active windows, and directed structure when the data are generated from the model family.

Theory anchor

This is the controlled support-recovery test for Theorem 3, while Theorem 1 supplies the separability conditions under which the active block being recovered is identifiable in the first place.

Architecture

```
Synthetic generator
-> event stream with planted active subgroup, hub, active windows, and marks
```

```
-> fit proposed model and baselines  
-> evaluate against known truth
```

Main outputs

- Member-set F1: did the model find the active actors?
- Hub accuracy: did the model identify the directional sender role?
- Active-event AUC: did the model localize the active regime in time?
- Exact support recovery where available.

Baselines

- **B1: Static Hawkes + gates.** Removes the latent regime field.
- **B2: Modular HMM + Hawkes.** Separates temporal segmentation from network assignment.
- **B3: Static Hawkes + SVD.** Tests whether time-averaging destroys episodic structure.
- **B4: Change-point Hawkes.** Tests whether a change-point-plus-Hawkes pipeline can replace joint switching inference.

B4 is included as an external-style two-step baseline rather than an internal ablation. It belongs in the synthetic baseline recovery layer because the synthetic setting provides clean ground truth for whether the change-point pipeline recovers the active subgroup.

4. Synthetic P2 true-alpha weakest-link sweep

Goal

Test the theorem-facing weakest-link coordinate directly: at fixed aggregate receiver-directed active mass, the weak receiver's true coupling `alpha_min` is controlled by the generator.

Theory anchor

This experiment targets the `alpha_min` term in Theorem 3: at fixed aggregate active evidence rate, the theorem predicts that the active-duration requirement grows quadratically as the weakest active coupling decreases.

Architecture

```
Synthetic star/hub generator  
-> designate one receiver as the weak link  
-> vary true alpha_min  
-> redistribute remaining active mass to keep aggregate active rate fixed  
-> sweep active duration pi1T  
-> fit proposed model  
-> evaluate weak-receiver recovery threshold
```

Interpretation

This is the only experiment used for P2. It validates the true `alpha_min` instrument and the expected difficulty ordering. In the current grid, only the strongest positive weak-link level crosses the 70% recovery threshold, while weaker positive levels are right-censored. Therefore, the run is reported as a censored weakest-link threshold diagnostic rather than as an empirical estimate of the `-2` exponent.

The zero-coupling row is a control for background false recovery. It is reported in tables but excluded from any log-log exponent fit because `log(0)` is undefined and because it represents no true weak active link.

5. Powered sparse directed-edge diagnostic

Goal

Show that the recovered directional score contains ordered-edge information, not only member-set or hub information.

Theory anchor

Theorem 1 identifies the active marked kernel block under the stated separability conditions; this diagnostic asks whether the estimated gauge-invariant directed scores retain ordered-edge information inside the recovered active block.

Architecture

```
Synthetic generator with larger active subgroup
-> plant sparse directed adjacency inside subgroup
-> fit proposed model
-> rank ordered non-self pairs by gauge-invariant score  $s_{\{j \rightarrow i\}}$ 
-> compare ranked scores to planted ordered edge set
```

Main outputs

- F1 at the planted edge count.
- Directed-edge AUC.
- Chance F1 based on planted edge density.
- Chance AUC = 0.5.

This test is deliberately harder than member-set recovery. It supports the directional low-rank machinery only if ordered-edge recovery is above chance.

6. C6 posterior-accuracy bridge

Goal

Test whether the E-step posterior is accurate enough to support the oracle-to-approximate transfer argument in the recovery theorem.

Theory anchor

This is the empirical diagnostic for Assumption Assumption 2, item 4 (Condition C6 in Appendix A.3). That condition is what allows Theorem 3 to transfer the oracle active-block support-recovery argument to the implemented generalized-EM estimator.

Architecture

```
Synthetic data with known true parameters
-> compute true-parameter oracle E-step posterior gamma*
-> compute fitted EM posterior gamma_hat
-> align labels globally
-> measure event-weighted posterior gap R_gamma
-> corrupt the posterior deliberately
-> clamp corrupted posterior and rerun recovery
-> observe which recovery metrics degrade
```

The reference object is the oracle posterior under the true parameters, not the hard latent path. The corruption experiment is the causal part: it directly tests how recovery changes as the posterior bridge is degraded.

Interpretation

Coarse member recovery can remain robust under some posterior error, while hub and directional recovery are more sensitive. This is exactly the role of the posterior-accuracy condition: it is most important for fine directional structure, not merely for finding generally active actors.

7. Design ablations D1-D4

Goal

Show that the model components are identifiability repairs, not arbitrary engineering choices.

Theory anchor

The ablations are tied directly to the paper's identifiability results. D1 tests the birth-time attribution requirement in Theorem 1; D2 tests sender/receiver role separation needed for active support recovery; D3 tests whether the active interaction field remains recoverable under a less structured parameterization; and D4 tests the birth-state-blind collinearity collapse in Theorem 2. Mark-channel normalization is tied to Proposition 1.

Architecture

```
Same synthetic benchmarks
-> fit variants that remove one repair at a time
-> compare member F1, hub accuracy, active-event AUC, and kernel diagnostics
```

Ablations

- **D1: Present-time gating.** Replaces birth-time gating with present-time gating. Tests whether excitation must be attributed to the parent event's birth regime.
- **D2: Single actor gate.** Collapses sender and receiver gates. Tests whether sender/receiver role separation is necessary.
- **D3: Dense active block.** Removes the low-rank active coupling structure. Tests whether a parsimonious active interaction field is needed.
- **D4: Shared beta/rho.** Enforces shared temporal decay and mark-channel structure across regimes. Tests temporal-shape and mark-channel separation.

D4 also has a separate kernel-identifiability diagnostic. This matters because support recovery can look plausible even when the active kernel itself is wrong.

8. UCDP GED planted coordination

Role in the evidence ladder

UCDP GED is the main real social-science background. It is used to test whether the exposure story survives on a real, heterogeneous event stream while preventing raw actor volume from solving the task.

Theory anchor

This is the real-background P1 test for Theorem 3: recovery should track active-born effective exposure N_{eff} , not total event volume or marginal actor frequency.

Architecture

```
UCDP GED raw events
-> preprocess exact-date conflict events
-> map selected dyad streams to actors
-> map fatality-severity bins to marks
-> construct volume-neutral planted coordination
-> sweep realized active-born exposure  $N_{eff}$ 
-> fit proposed model and baselines
-> evaluate member F1, receiver F1, active AUC, and exact recovery diagnostics
```

Interpretation

The UCDP design is a **volume-neutral planted coordination**, because the UCDP design is stronger than a simple additive injection. Marginal actor volume is not sufficient to identify the planted subgroup.

UCDP directly supports P1 when recovery improves with realized planted active-born exposure while baseline recovery remains near the chance-scale floor. UCDP is not used for the P2 exponent fit because the sharp quadratic weakest-link prediction requires a controlled true-alpha synthetic sweep. UCDP can provide hard-path posterior diagnostics, but it is not a theorem-clean P3 test without a true-parameter oracle posterior.

9. CRCNS CA1 planted assembly and raw CA1 predictive checks

Raw CA1 role

Raw CA1 has no planted active support. It is used for label-free predictive checks and structural diagnostics. The theory anchor is Theorem 1: the raw fit can only be read as evidence that the model finds useful separated latent structure, not as proof of true latent coordination.

Raw CRCNS CA1 events

- > retain top K units
- > single mark planted assembly stream
- > contiguous train/test split
- > fit proposed model and comparable baselines
- > evaluate held-out NLL/event

The raw CA1 likelihood experiment asks whether the structured switching model explains held-out neural event streams better than static alternatives. It does not prove real latent coordination because there are no labels.

Planted assembly role

The planted assembly experiment is the real-background exposure analogue of Theorem 3, while the modular B2 failure tests the segmentation-without-assignment failure mode implied by the joint identifiability framing.

Real CA1 background

- > add known four-unit planted assembly and active windows
- > sweep injection strength / realized active-born events
- > fit proposed model and baselines
- > evaluate planted assembly recovery

The paper-facing term is **CRCNS CA1 planted assembly**. The planted assembly experiment restores ground truth on a realistic neural background. It is additive rather than volume-neutral, so it is treated as a real-background replication of the exposure pattern rather than the cleanest theorem-aligned exposure test.

The key P4 interpretation is segmentation without assignment: B2 can locate active intervals but still assign the active actors incorrectly.

10. FTX planted receiver cascade and mark-channel stress test

Raw FTX role

Raw FTX windows have no planted active support. They are used for held-out predictive checks and temporal/structural diagnostics around a real market-stress window. The theory anchor is conditional: without planted labels or an oracle posterior, raw FTX can support predictive plausibility but cannot verify Theorem 3.

Binance aggregate trades

- > filter large notional trades
- > map assets to actors
- > map taker side to marks
- > slice two-hour windows

- > fit proposed model and baselines
- > evaluate held-out NLL/event and active-structure diagnostics

Planted receiver cascade role

The planted receiver cascade tests the directed receiver side of the active block identified in Theorem 1 under a hard real market background.

FTX peak background

- > plant FTX-rooted receiver cascade into selected assets
- > fit proposed model
- > evaluate target receiver recovery and source retention

The paper-facing term is **FTX planted receiver cascade**. This is a targeted recovery probe on a hard, nonstationary real market background. It is not a causal analysis of the FTX collapse.

Mark-channel role

The mark-channel test is tied to Proposition 1 and the mark-normalization part of Theorem 1: actor couplings and mark-transition channels must be separated to recover a meaningful marked active block.

Buyer/seller trade marks

- > plant marked cascade
- > fit marked model with ρ updates
- > compare active and dormant mark-transition matrices $\rho^{(1)} - \rho^{(0)}$

This is the cleanest test of the marked-process claim. An unmarked model can represent volume and actor activity, but it cannot represent a sell-to-buy or buy-to-sell propagation channel. The FTX mark-channel experiment checks whether the active marked block recovers a planted cross-mark propagation signature.

11. Raw predictive checks

Raw CA1 and raw FTX are label-free. Their architecture is:

Real event stream

- > contiguous train/test split
- > fit proposed model and comparable baselines on train
- > evaluate latent-CTMC-marginal held-out NLL/event on test

Only methods sharing the same marked-event likelihood scale are treated as directly comparable. The modular HMM baseline has an interval-count likelihood and is interpreted as a diagnostic rather than a like-for-like NLL competitor.

12. How to read the evidence

The empirical section triangulates the same claim from multiple directions:

1. Known-truth synthetic recovery when the identifiable structure in Theorem 1 is present and the exposure conditions in Theorem 3 are favorable.
2. Controlled collapse when a required condition is removed, including the D4 birth-state-blind collapse predicted by Theorem 2.
3. Posterior-accuracy stress tests showing that Assumption Assumption 2, item 4 / Condition C6 matters for fine directional recovery.
4. Volume-neutral planted coordination on UCDP GED as a real social-science exposure test for **Neff** in Theorem 3.
5. Planted assembly recovery on CRCNS CA1 as a real neural-background replication of the exposure pattern.
6. Raw CA1 and FTX predictive checks without planted labels, interpreted only as label-free plausibility diagnostics.
7. FTX planted receiver cascade and mark-channel recovery as a cross-domain stress test for the directed and marked active block in Theorem 1 and Proposition 1.

The intended interpretation is not that every dataset proves every theorem condition. Instead, each layer has a specific evidential role. Synthetic data support theorem-clean claims; real-background planted experiments test robustness against realistic burstiness; raw real-data fits support predictive plausibility; and ablations show that the model's components correspond to identifiable failure modes.

13. Connection to commands

The README contains the executable commands. This architecture file explains why those commands exist and how their outputs map to the paper's evidence layers.

The main command families are:

- Synthetic baseline recovery, including B4 base synthetic.
- Synthetic design ablations D1-D4.
- Synthetic P2 true-alpha weakest-link sweep and evaluator.
- Synthetic C6 posterior-accuracy diagnostics.
- Powered sparse directed-edge diagnostic.
- UCDP GED preprocessing, window screening, volume-neutral planted coordination, and weak-link diagnostic.
- CRCNS CA1 raw processing, planted assembly construction, held-out likelihood, permutation diagnostic, and planted assembly recovery.
- FTX preprocessing, temporal windows, planted receiver cascade, mark-channel rho recovery, and optional lag-correlation diagnostic.